When May an Agent Claim Success? Supplementary Material

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1 CANONICAL CONTRACT

A Verification Contract serializes the tuple $K = \langle I, P, B, V, A, R \rangle$. The released JSON Schema requires immutable identifiers and versions, at least one typed proof obligation, task and policy binding, a three-valued decision space, an admissible-transition allowlist, and a receipt-schema version. Obligations and observations are separate objects: observations cannot introduce a new acceptance criterion after evidence submission.


```
{
 "contract_id": "k-demo", "version": 1,
 "proof_obligations": [{
   "obligation_id": "content", "required": true,
   "evidence_type": "image",
   "predicate": "contains required statement",
   "failure_mode": "reject"
 }],
 "binding": {"task_id": "t-demo",
   "policy_version": 1, "hash_algorithm": "sha256"},
 "verifier": {"decision_space":
   ["accept", "reject", "abstain"],
   "threshold_version": "theta-1"},
 "admissible_transitions": ["mark_complete"]
}
```


2 THREE-VALUED AUTHORIZATION

Every authorization predicate evaluates to true, false, or unknown. Unknown never coerces to true. Required predicates all true imply accept; any known failed obligation implies reject; otherwise the decision is abstain. Positive transition eligibility additionally requires the action to appear in the contract allowlist and the verifier not to be in a declared degraded state.

Table 1: Decision semantics.

Required states	Decision	Positive transition
all true	accept	allowlisted action only
any false	reject	blocked
otherwise/unknown	abstain	blocked

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3 VERSION AND AMENDMENT PROTOCOL

The first evidence commitment freezes K_v. Any policy change creates K_{v+1} with a semantic diff, initiator, reason, timestamp, and evidence-invalidation rule. Evidence bound to K_v is inadmissible under K_{v+1} unless both versions explicitly permit reuse. The original contract and receipt remain immutable.

4 CANONICAL RECEIPT

A receipt contains contract, task, policy, and receipt identifiers; evidence digests; per-obligation states and reason codes; provider and threshold versions; degradation state; final decision; transition action and eligibility; and appeal linkage. Machine completeness and human reconstruction are distinct metrics. Appeals append an adjudication event rather than rewriting history.

5 CONTROLLED DEFECTS

6 METRICS AND ANTI-GAMING CHECKS

Let N be all cases, A automated cases, C accepted cases, and N^- adjudicated-negative cases. We report total automation coverage $|A|/N$, acceptance coverage $|C|/N$, unsafe acceptance among accepted cases, unsafe acceptance per all cases, and unsafe acceptance per negative case. Reporting acceptance coverage prevents a system from appearing safe by satisfying total coverage mainly through rejection.

Selective loss weights a false acceptance five times a correctable false rejection in the confirmatory analysis. Sensitivity at ratios 1:1, 2:1, and 10:1 is secondary. Risk–coverage curves and AURC are reported alongside the frozen 70% operating point.

7 STATISTICAL PROTOCOL

The base case is the resampling unit. Ten thousand clustered bootstrap resamples preserve dependence among matched variants. The primary estimate is the paired K-Full minus J1 unsafe-positive-transition difference. Raw counts, risk difference, risk ratio, and 95% intervals are mandatory. With too few independent clusters, a paired randomization analysis is reported as a robustness check. Secondary tests use Holm correction; model families are reported separately before pooling.

8 HUMAN STUDIES

The review experiment compares raw task/evidence against a contract packet without the final verdict. Primary outcome is unsafe acceptance; secondary outcomes are accuracy, time, confidence, workload, agreement, and confidence calibration. The audit experiment compares a canonical receipt with an unstructured application log and asks participants to recover policy version, decisive obligation, evidence commitment, degradation path, and transition eligibility.

Table 2: Mutation families and expected safe behavior.

Family	Controlled defect	Target property	Expected behavior
Policy	deadline/criterion changed after submission	ex-ante consistency	block or versioned amendment
Binding	artifact replayed across task/version	evidence binding	reject or abstain
Binding	stale artifact outside capture window	evidence binding	reject
Decision	provider missing or mandatory modality unsupported	selective safety	abstain
Decision	hard integrity conflict	selective safety	abstain
Receipt	policy version or decisive observation absent	reconstructability	transition blocked

9 PRIVACY AND GOVERNANCE

Raw artifacts remain access-controlled. Public records contain commitments, versions, derived observations, and privacy-scoped references. Requested fields, submitted fields, PII-bearing fields, retained bytes, and decision-irrelevant fields are measured. Legal, medical, employment, credit, biometric, intimate, and covert-surveillance cases are excluded.

10 REPRODUCIBILITY GATES

Submission requires a consented confirmatory manifest, sealed independent labels, immutable outputs, real-result metrics, provenance hashes, regenerated tables, privacy and mutation-leakage audits, synthetic isolation, and clean-environment reproduction. Synthetic dry runs validate code paths but cannot satisfy an empirical gate.

11 CLAIM BOUNDARIES

The protocol does not prove external-world truth, universal media authenticity, authorship, policy legitimacy, unbiased human review, or global safety of long-horizon workflows. It establishes only protocol eligibility under declared predicates.